

# ContestBench: Do Medical LLMs Know When Physicians Disagree?

Anonymous submission

## Abstract

Every widely used benchmark for medical large language models assumes each clinical question has a single correct answer. For a non-trivial fraction of real medicine this is false: equally qualified physicians disagree, and that disagreement is documented and clinically meaningful, not noise. A model that answers a contested case with high confidence invites over-reliance exactly where independent judgment is most needed, yet accuracy- and calibration-based benchmarks cannot detect it, since they score confidence against a gold label the case does not possess. We introduce **ContestBench**, which replaces the binary gold label with the empirical distribution of physician opinion. Using LIDC-IDRI, where four radiologists independently rate each lung nodule, we derive a physician *agreement rate*  $\pi \in [0.5, 1]$  per case and score confidence by **Physician Agreement Deviation (PAD)**, the L1 analogue of the Brier proper scoring rule applied to a distributional target. We evaluate ten model/reasoning configurations across three vendors on the full corpus ( $n=1,413$ ) under one standardized elicitation. Confidence is linearly uncorrelated with agreement (Pearson  $r$  near zero everywhere) and only weakly, inconsistently non-linearly dependent on it. A **case-blind constant** (always 0.75) achieves PAD 0.143, and **all ten configurations are significantly worse**. This is not because difficulty is unpredictable: a predictor trained on the same rating-independent features the models see attains  $\text{PAD} \approx 0.127$ , a **vignette-optimal** target no model reaches. A post-hoc recalibrator mapping a model’s own confidence plus four cheap features onto  $\pi$  closes almost the entire gap for every configuration; an ablation shows confidence is statistically redundant with those features for this purpose, though a small independently-detectable signal survives. Extended reasoning does not help (a mixed-sign null across five families), accuracy-based calibration *worsens* PAD for every configuration, in-context prompting closes the gap only for the smallest model, and standard Expected Calibration Error is uncorrelated with PAD and degenerate on exactly the cases the benchmark targets. Finally, a reliance simulation shows PAD-minimizing recalibration *worsens* simulated over-reliance for models that were already appropriately cautious, illustrating that optimizing an average calibration metric is not the same as minimizing a safety-relevant downstream harm. ContestBench and PAD give the field a measurement instrument for a

failure mode existing evaluation cannot see: LLM confidence decoupled from clinical difficulty.

## 1 Introduction

**A hidden assumption.** The benchmarks that govern medical LLM development, MedQA (Jin et al. 2021), MedMCQA (Pal, Umapathi, and Sankarasubbu 2022), and their descendants (MedR-Bench Contributors 2025; MedThink-Bench Contributors 2025; Li et al. 2024), share an invisible premise: every clinical question has one correct answer against which a model can be scored. This is false for a meaningful share of clinical medicine. Inter-physician disagreement is extensively documented and is not measurement error; it reflects genuine evidentiary uncertainty and legitimate variation in expert judgment. Two equally experienced radiologists can read the same chest CT and reach opposite conclusions because the case is genuinely ambiguous, not because one is wrong.

**Why it matters.** A model expressing high confidence on a genuinely contested case fails in a specific, predictable way: it drives over-reliance. A model that is uniformly confident regardless of difficulty teaches users to either discount its confidence entirely or defer on precisely the cases where independent judgment matters most – the automation-complacency failure mode documented in aviation and decision-support research (Parasuraman and Riley 1997). No current evaluation can detect this: accuracy benchmarks reward the confident answer matching the gold label, and calibration metrics such as Expected Calibration Error (ECE) require a binary correct/incorrect outcome that a split physician panel does not provide. On exactly the cases needing scrutiny, the standard tools are undefined or degenerate.

**Our approach.** We measure confidence against the *distribution* of expert opinion rather than a single label, building ContestBench on LIDC-IDRI (Armato et al. 2011), where four radiologists independently rate each nodule’s malignancy. From these ratings we compute a per-case physician agreement rate  $\pi = \max(f, 1 - f) \in [0.5, 1]$ ,  $f$  being the fraction rating the nodule malignant. A well-behaved model should be confident when physicians agree ( $\pi$  near 1) and hedge when they split ( $\pi$  near 0.5). We quantify deviation from this ideal with **Physician Agreement Deviation**,

---

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

$PAD = \text{mean}|c - \pi|$ , the L1 form of the Brier proper scoring rule (Brier 1950; Gneiting and Raftery 2007) applied to a distributional target. Every model is evaluated through one standardized elicitation at full corpus scale ( $n=1,413$ ), after confirming that absolute PAD is elicitation-sensitive and cross-model comparison therefore requires it held fixed.

**What we find.** Model confidence does not track physician agreement (correlation indistinguishable from zero; only half of configurations show an MI excess surviving multiple-comparison correction). A **case-blind constant** beats every one of ten configurations significantly; a **vignette-optimal oracle** using only features already in the models’ prompts reaches  $PAD \approx 0.127$ , so the failure is one of use, not access; **extended reasoning does not close the gap** (mixed-sign,  $|\Delta PAD| \leq 0.033$  across five families, including a DeepSeek-R1-lineage model); and **ECE cannot see any of this**. A mechanism analysis rules out the benign story that confidence is noise: models respond substantially to case features ( $R^2$  up to 0.76) but their feature-to-confidence mappings contradict one another, and models do not even agree on which cases are hard.

**Contributions.** (1) **ContestBench**, scoring medical-LLM confidence against a documented physician-agreement distribution rather than a gold label, on a public, IRB-free corpus. (2) **PAD**, a proper-scoring-rule-motivated metric for a distributional target, with a demonstration that ECE is degenerate exactly where the benchmark is informative and that accuracy-calibration does not imply agreement-calibration. (3) A **full-scale, ten-configuration evaluation** showing the decoupling is universal, that a cheap oracle beats every model, and that reasoning and prompting do not close the gap. (4) A **post-hoc recalibrator** that closes the oracle gap for every model, paired with an over-reliance simulation showing this metric-optimal fix is not automatically harm-optimal. (5) A **reproducible pipeline** regenerable end to end from raw annotation files.

## 2 Related Work

**Single-label medical QA.** MedQA (Jin et al. 2021) and MedMCQA (Pal, Umapathi, and Sankarasubbu 2022), their reasoning-focused successors (MedR-Bench Contributors 2025; MedThink-Bench Contributors 2025), and adaptive frameworks such as MEDIQ (Li et al. 2024) all score answers against a single correct choice. ContestBench targets exactly the population – genuinely contested cases – for which this premise is false, asking not whether the model is right but whether its confidence tracks how right an answer *can be*.

**Learning from disagreement.** A parallel NLP line argues a single ground truth discards annotator-disagreement information and proposes training on soft labels instead (Davani, Díaz, and Prabhakaran 2022; Uma et al. 2021). ContestBench reuses this insight for a different purpose: a *deployment-time evaluation instrument* with proper-scoring-rule semantics, on annotators who are practicing physicians.

**Calibration in medicine and beyond.** ECE and its binning variants (Guo et al. 2017; Naeini, Cooper, and Hauskrecht 2015) and Platt scaling (Platt 1999), which we

use as an accuracy-calibration baseline, are all anchored to a single correct answer per case and are undefined for a split panel. Byun et al. (2026) study overconfidence and Platt-scaling in medical VQA; Savage et al. (2025) use sample-consistency as a non-verbalized signal (reused here as a robustness check, Section 5.11); Testoni and Calixto (2026) study specialty-aware clinical-QA uncertainty. Our accuracy-calibration baseline (Section 5.10) makes the distinction concrete: calibrating to accuracy *worsens* PAD for every configuration.

**Verbalized LLM confidence.** Kadavath et al. (2022) and Tian et al. (2023) study whether LLMs’ stated confidence is well-calibrated to correctness; Xiong et al. (2024) benchmark elicitation strategies, Lin, Hilton, and Evans (2022) teach models to express uncertainty in words, Mielke et al. (2022) reduce conversational overconfidence via linguistic calibration, Ovadia et al. (2019) evaluate predictive uncertainty under distribution shift more broadly, and Jiang et al. (2021); Desai and Durrett (2020) study calibration of pre-trained transformers and QA models. Steyvers et al. (2025) additionally show a mismatch between what LLMs know and what users think they know. All target correctness calibration or human perception of it; none address a distributional target. Singhal et al. (2023) and clinical LLM evaluations motivate our domain choice, and Kompa, Snoek, and Beam (2021) motivate uncertainty communication in medical ML as a deployment concern, which we operationalize via the over-reliance simulation (Section 5.12), itself building on an HCI literature showing AI explanations and confidence signals do not reliably prevent overreliance in practice (Bansal et al. 2021; Bućinca, Malaya, and Gajos 2021; Passi and Vorvoreanu 2022).

**Concurrent work.** MedQADE Authors (2026), concurrent with this work, finds physicians scale caution with difficulty while LLM judges evaluating open-response answers do not – complementary evidence from the judge/evaluator side rather than the diagnostician side, in a different domain and design (accuracy-anchored judge evaluation vs. our proper-scoring-rule evaluation of the model as diagnostician).

**Prior work and scope.** PAD is the L1 analogue of the Brier score (Brier 1950), itself a strictly proper scoring rule (Gneiting and Raftery 2007) in the forecast-evaluation tradition of Murphy (1973) and DeGroot and Fienberg (1983). That agreement is partially predictable from case features is an unsurprising fact, not a claim requiring novel methodology; our oracle and recalibrator (Sections 4.3, 4.4) reuse it diagnostically, not as a novel prediction method. Our over-reliance simulation (Section 5.12) is anchored in Lee and See (2004) and Parasuraman and Riley (1997).

## 3 The ContestBench Corpus

### 3.1 Physician Agreement from LIDC-IDRI

LIDC-IDRI (Armato et al. 2011) is a public lung-CT corpus in which up to four radiologists independently annotate each nodule, including a five-point malignancy rating. No DICOM imagery is available; we parse the annotation XML directly, extracting each reader-nodule’s ROI polygon and its

malignancy, subtlety, spiculation, and margin ratings. Since a raw `noduleID` is not globally unique across a scan’s four reading sessions, we match nodules by XML-native centroid proximity (in-plane  $< 30\text{px}$ , slice  $< 3\text{mm}$ ; frozen after confirming corpus size and tier composition are stable within  $\pm 5\%$  across a  $2\times$  range of thresholds, Appendix B), retaining groups with  $\geq 3$  readers:  $n=1,413$  nodules across 709 scans.

We define  $f$  as the fraction of readers rating malignancy  $\geq 3$  of 5 (the LIDC-IDRI field convention), and agreement rate  $\pi = \max(f, 1 - f) \in [0.5, 1]$  – the fraction agreeing with the majority side.  $\pi$ , not  $f$ , is the correct target: comparing confidence to  $f$  directly is a category error, since a correctly-confident model on a unanimous-benign case ( $f=0$ ) would register a spurious deviation.  $\pi$  is discrete under 4-reader support ( $\{.5, .75, 1\}$ ; 3-reader nodules add the one new value  $\{.667\}$ ), so we tier cases as **high** ( $n=546$ ), **contested** ( $n=708$ ), and **ambiguous** ( $n=159$ , structurally all exact 4-reader splits). An exclusion audit (Appendix B) confirms drops are almost entirely LIDC’s inherent sub-3mm/uncharacterized marks. Appendix C separately shows the baseline-beating result is robust to the  $\geq 3$  threshold choice, while the pooled correlation is not – fully decomposed there, not left as an open question.

### 3.2 The Vignette Template and Its Collisions

Each case is presented as a vignette reporting only mean subtlety, spiculation, and margin (Section 4.2). This coarseness has a consequence we treat as core method, not a footnote: 1,413 cases collapse to 690 distinct vignettes, and **54.8%** ( $n=775$ ) share an identical vignette with a different- $\pi$  case (mean  $\pi$ -spread 0.30), which the model cannot distinguish by construction. We report the achievable target as **vignette-optimal** rather than a model-agnostic ceiling (Section 4.3), and confirm on the collision-free 45.2% subset ( $n=638$ ) that decoupling is not a collision artifact (Section 5.2). Appendix A gives worked example vignettes, including a colliding pair.

## 4 Method

### 4.1 Physician Agreement Deviation (PAD)

For confidence  $c \in [0, 1]$  and agreement  $\pi \in [0.5, 1]$ ,

$$\text{PAD} = \frac{1}{n} \sum_i |c_i - \pi_i|, \quad \text{PAD}^+ = \frac{1}{n} \sum_i (c_i - \pi_i),$$

unsigned PAD as the headline statistic and signed  $\text{PAD}^+$  as a directional diagnostic by tier ( $\text{PAD}^2$  reported where noted). **Claim (unverified proof).** We claim, without independent proof verification, that PAD inherits a proper-scoring-rule guarantee: if  $\pi$  is the calibrated probability a random reader agrees with the majority, expected PAD is minimized at  $c = \pi$ . We flag this, not the empirical results, as the paper’s one unconfirmed theoretical component.

### 4.2 Elicitation and Provenance

Every model receives the identical instruction: a Malignant/Benign answer and a 0–100 confidence for a nodule described only by mean subtlety, spiculation, and margin,

as one JSON object. An earlier pilot mixed pipe-delimited and JSON formats across models; we caught this before it became load-bearing and standardized on one elicitation for the reported panel. We evaluate ten configurations (Anthropic Haiku/Sonnet/Opus, Google Gemini, DeepSeek, each standard/thinking); Anthropic and Gemini were collected via logged APIs (model string, temperature where settable, prompt hash per call), while DeepSeek is a documented manual-collection exception (no usable API path). **Provenance correction, disclosed in full:** a manual, uncontrolled-temperature pass was later re-collected via API; Haiku’s PAD rose substantially (0.16→0.35–0.36) because the manual interface had deceptively clustered confidence near the corpus-optimal constant, while every other configuration was stable within hundredths of PAD across both routes. We report only corrected, API-provenanced values; this strengthens rather than weakens the central claim, since no substantive conclusion depends on the one configuration that looked deceptively well-calibrated under noisier collection. Gemini:standard reached 96% coverage ( $n=1,350$ ) after key-rotation retries – a deliberate stop, PAD stable at 0.178–0.180 across every coverage level tried. All other configurations reached  $n \geq 1,412$ ; manual and API artifacts and per-call provenance logs are preserved.

### 4.3 Vignette-Optimal Oracle

To measure recoverable signal, we fit a 5-fold out-of-fold predictor of  $\pi$  at three feature levels: **geometry-only** (ROI extent, the only feature with zero subjective input), **rating-independent** (subtlety, spiculation, margin, extent – independent of the malignancy rating specifically, and exactly what the LLMs see), and **all-features** (adds reader count and malignancy-extremity, which share source with  $\pi$ ). We check gradient-boosted (200 estimators, depth 3, lr 0.05), ridge ( $\alpha=1$ ), and linear estimators. This is the **vignette-optimal** target – the best performance achievable from the same coarse, collision-prone text the models see (Section 3.2), not a model-agnostic bound.

### 4.4 Post-Hoc Recalibrator

We separately fit a per-configuration recalibrator mapping [model confidence + the four rating-independent features] onto  $\pi$ , same out-of-fold protocol. Comparing oracle (features only) against recalibrator (features + confidence) on identical folds gives a bootstrap CI on confidence’s marginal contribution (Section 5.5). Every stochastic procedure here – CV splits, bootstrap, permutation nulls – uses a fixed code-level seed; we report draw counts in prose and release the seeded code.

## 5 Results

### 5.1 No Model Beats a Case-Blind Constant

The PAD-minimizing constant is  $c^*=0.75$ ,  $\text{PAD}=0.143$ . Table 1 reports, per configuration,  $n$ , mean confidence,  $r(c, \pi)$ , PAD, and the bootstrap-CI’d gap to  $c^*$ . Every gap’s 95% CI is strictly positive: **all ten configurations are significantly worse than never looking at the case.** The best is Gemini:standard (PAD 0.178), indistinguishable

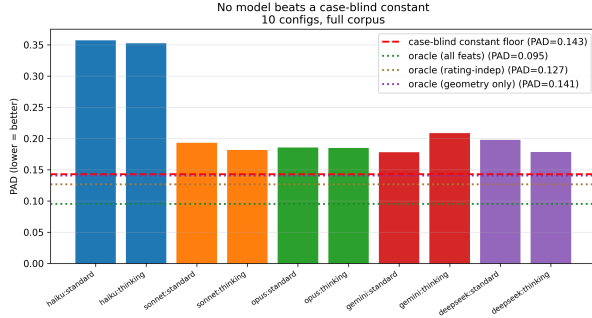


Figure 1: PAD per configuration against the case-blind constant floor (red) and the vignette-optimal oracle (dashed). No configuration reaches the constant.

from DeepSeek:thinking (0.179); an earlier, uncorrected pass had reported Haiku:thinking (0.157) as best, superseded by the provenance correction (Section 4.2), which strengthens rather than weakens this claim. Figure 1 shows every PAD against the constant and oracle.

## 5.2 Confidence Is Decoupled from Agreement

Correlations are uniformly near zero ( $-0.078$  to  $+0.258$ , Table 1), with confidence essentially flat while  $\pi$  ranges over  $[0.5, 1]$ . A mutual-information test against a 1,000-permutation null ( $k=3$  kNN, Table 2) shows 8/10 configurations exceed their null’s 95th percentile, but only 5/10 survive Bonferroni ( $\alpha=0.005$ ); two (haiku:standard, sonnet:standard) show no excess at  $k=3$ . This split is itself  $k$ -sensitive: at  $k=5, 10$ , three more configurations become significant, while sonnet:standard and gemini:standard remain borderline at every  $k$  ( $p \in [0.004, 0.02]$ ). The defensible statement: confidence is linearly uncorrelated with agreement and only weakly, inconsistently non-linearly dependent on it – not flatly independent, but not useably predictive. The oracle’s own MI (0.118) exceeds every model’s. Restricting to the collision-free 45.2% subset ( $n=638$ ) reproduces the null (pooled  $r=+0.07$ ), so this is not a collision artifact.

The pooled near-zero correlation is also not uniform once cases are split by sensitivity to the malignancy-positive threshold defining  $\pi$ : on the  $\approx 61\%$  of cases whose tier flips between  $\geq 3$  and  $\geq 4$ , the correlation’s *sign* depends on which threshold is used (a composition artifact of the redefinition, confirmed via the decomposition in Appendix C, not a change in model output); on the complementary  $\approx 39\%$  stable subset, four of five checked configurations show a real, threshold-invariant *positive* correlation ( $+0.20$  to  $+0.25$ ) that the pooled statistic averages away. We report the pooled  $\geq 3$ -threshold statistic as primary because  $\geq 3$  is the field-standard convention, but flag this as a genuine refinement, not a footnote.

## 5.3 The Vignette-Optimal Oracle Beats Every Model

Table 3 reports oracle PAD across feature levels and estimators. Rating-independent (features already shown to every

LLM) attains  $\text{PAD} \approx 0.127\text{--}0.133$  robustly, beating every model (best 0.178). The all-features headline (0.095) is GBT-specific (ridge/ linear: 0.131); geometry alone (0.139–0.141) is indistinguishable from the constant, so the signal lives in subjective Likert calls, not raw size. A leave-one-out bucket-mean baseline (no fitting) gives 0.140 (indistinguishable from the constant); in-sample is 0.067 but leaky. **Conclusion:** the vignette affords only 0.010–0.016 PAD of usable margin over ignoring the case, depending on estimator, real and achievable from the same prompt the models see – and no model captures any of it.

## 5.4 Mechanism: What Drives Confidence Instead

Confidence is not noise: OLS on the locked feature set attains  $R^2$  from 0.054 (haiku:standard) to 0.758 (deepseek:standard). Haiku’s low  $R^2$  is consistent with its provenance-corrected confidence being noisier, a data-quality artifact of that configuration, not evidence the analysis is compromised elsewhere. Feature-to-confidence mappings *contradict* across models (e.g., the spiculation coefficient is strongly positive in one configuration, strongly negative in another), and pairwise cross-model confidence correlation spans  $-0.30$  to  $+0.84$  (mean  $+0.27$ ): models do not agree on which cases are hard, ruling out a shared latent difficulty signal and pointing instead to idiosyncratic, model-specific heuristics.

## 5.5 Post-Hoc Recalibration and the Confidence Ablation

Table 4 shows the recalibrator closes essentially the entire oracle gap for every configuration (raw PAD 0.178–0.357  $\rightarrow$  0.125–0.127, matching the oracle’s 0.127), and gives the ablation: comparing features-only against features+confidence on identical folds, mean gap  $+0.0007$  PAD, with 8/10 configurations’ bootstrap CI bracketing zero. Two (haiku:standard, sonnet:standard) exclude zero, but by  $\approx 0.0016$  PAD, an order of magnitude below any other effect in this paper. Held-out permutation importance of confidence is small but non-zero for most configurations (mean 0.023, range 0.004–0.064). **Non-contradictory claim:** confidence is statistically redundant with four cheap features for this purpose – its marginal PAD contribution is indistinguishable from zero for most configurations, negligible even where technically significant – while still carrying a small, real, importance-detectable signal that is almost entirely redundant with those features. The recalibrator succeeds by routing *around* confidence, not extracting from it.

## 5.6 Prompting Does Not Fix It

We test whether in-context exemplars shift confidence toward  $\pi$  (three designs: A, an explicit rule; B, a qualitative hedge; C, worked examples; crossed with Haiku/Sonnet/Opus, on a shared, leak-free 300-case held-out set). Table 5: **Haiku improves under all three** ( $\text{PAD } 0.356 \rightarrow 0.141\text{--}0.168$ ), reaching statistical parity with the constant under B/C – the one case where an intervention reaches (not exceeds) the naive baseline. **Sonnet and Opus are largely inert or worse:** their best design (B) narrows the gap ( $\text{PAD } 0.158$ ) but remains significantly worse than the constant; Opus is actively hurt by

| config                         | $n$  | conf | $r(c, \pi)$ | PAD   | gap vs. constant [95% CI] |
|--------------------------------|------|------|-------------|-------|---------------------------|
| haiku:standard                 | 1412 | 0.53 | +0.042      | 0.357 | +0.214 [+0.203, +0.227]   |
| haiku:thinking                 | 1413 | 0.52 | +0.083      | 0.352 | +0.210 [+0.197, +0.223]   |
| sonnet:standard                | 1413 | 0.67 | -0.078      | 0.193 | +0.050 [+0.045, +0.055]   |
| sonnet:thinking                | 1413 | 0.71 | -0.041      | 0.182 | +0.039 [+0.034, +0.043]   |
| opus:standard                  | 1413 | 0.68 | -0.076      | 0.186 | +0.043 [+0.038, +0.048]   |
| opus:thinking                  | 1413 | 0.72 | -0.069      | 0.185 | +0.042 [+0.037, +0.047]   |
| gemini:standard                | 1350 | 0.79 | -0.078      | 0.178 | +0.035 [+0.029, +0.040]   |
| gemini:thinking                | 1413 | 0.66 | +0.071      | 0.209 | +0.066 [+0.057, +0.075]   |
| deepseek:standard <sup>†</sup> | 1413 | 0.64 | +0.258      | 0.198 | +0.055 [+0.049, +0.061]   |
| deepseek:thinking <sup>†</sup> | 1413 | 0.70 | -0.065      | 0.179 | +0.036 [+0.032, +0.040]   |

Table 1: Full-corpus panel results.  $c^* = 0.75$ ,  $\text{PAD}_{c^*} = 0.143$ . All ten gaps significant. <sup>†</sup>: manual-collection route; others API-provenanced.

| config         | MI    | excess | $p$   | Bonf.      |
|----------------|-------|--------|-------|------------|
| haiku:std      | 0.026 | -0.008 | 0.141 | no         |
| haiku:think    | 0.025 | +0.001 | 0.045 | no         |
| sonnet:std     | 0.013 | -0.013 | 0.189 | no         |
| sonnet:think   | 0.058 | +0.030 | 0.001 | <b>yes</b> |
| opus:std       | 0.057 | +0.020 | 0.001 | <b>yes</b> |
| opus:think     | 0.038 | +0.009 | 0.020 | no         |
| gemini:std     | 0.042 | +0.015 | 0.007 | no         |
| gemini:think   | 0.123 | +0.089 | 0.001 | <b>yes</b> |
| deepseek:std   | 0.095 | +0.054 | 0.001 | <b>yes</b> |
| deepseek:think | 0.091 | +0.056 | 0.001 | <b>yes</b> |
| oracle         | 0.118 | +0.092 | 0.001 | -          |

Table 2: MI(confidence,  $\pi$ ),  $k=3$ , vs. permutation null; excess = MI - null 95th pct. Bonferroni  $\alpha=0.005$ , 5/10 survive.

| oracle variant            | gbt   | ridge | linear |
|---------------------------|-------|-------|--------|
| geometry-only             | 0.141 | 0.139 | 0.139  |
| rating-independent        | 0.127 | 0.133 | 0.133  |
| all-features <sup>†</sup> | 0.095 | 0.131 | 0.131  |
| bucket-mean (LOO)         |       | 0.140 |        |
| case-blind constant       |       | 0.143 |        |
| best evaluated model      |       | 0.178 |        |

Table 3: Oracle PAD, 5-fold out-of-fold. <sup>†</sup>: shares source with  $\pi$ ; 0.095 is GBT-specific.

C (0.184  $\rightarrow$  0.204). No combination significantly beats the constant; prompting is not a fix, the recalibrator is.

## 5.7 Reasoning Effect: A Null Result

Toggling thinking changes PAD by a small amount everywhere; unlike every other comparison here, we give paired bootstrap 95% CIs rather than points alone: haiku -0.005 [-0.014, +0.004] (ns), sonnet -0.011 [-0.015, -0.008] (sig.), opus -0.001 [-0.004, +0.003] (ns), gemini +0.033 [+0.024, +0.043] (sig.), deepseek -0.019 [-0.025, -0.013] (sig.). Three of five show a detectable effect, but direction is inconsistent and every estimate is far smaller than any other effect in this paper. Reason-

ing neither fixes nor meaningfully worsens the decoupling, across five families including a DeepSeek-R1-lineage model.

## 5.8 Capability Trend (Suggestive)

Within the Anthropic family, signed PAD on the ambiguous tier trends upward with capability rank – more-capable models are more confidently wrong on exactly the cases physicians split on. We report this as suggestive only (three rungs, non-monotonic, Opus-driven); full numbers in Appendix D.

## 5.9 ECE Cannot See the Problem

ECE and PAD are near-uncorrelated: gpt-oss-20b – a supplementary, differently-elicited model (single-call JSON, not the ten-panel’s pipe-batch format, so not directly comparable to Table 1) – has excellent accuracy-calibration (ECE=0.050) alongside poor agreement-calibration (PAD=0.170, worse than the constant). ECE is additionally degenerate on the ambiguous tier, requiring a binary label a 50/50 split does not provide: assigning that label at random across 2,000 draws swings ECE across a 0.15-wide 95% band, while PAD needs no label and is a single stable value.

## 5.10 Accuracy-Calibration Is Not Agreement-Calibration

Platt-scaling each configuration’s confidence to its own accuracy (5-fold CV, no leakage) and re-measuring PAD against  $\pi$  *worsens* PAD for **every configuration**: pooled mean PAD rises 0.222  $\rightarrow$  0.368. Calibrating to a binary correctness label, the object it is designed for, provably cannot calibrate confidence to physician agreement.

## 5.11 Robustness Check: Non-Verbalized Confidence

On a reduced spot-check (two models,  $n=56-85$ ), a genuinely non-verbalized signal in the spirit of self-consistency and consistency-based hallucination detection (Wang et al. 2023; Manakul, Liusie, and Gales 2023) – sample-consistency across repeated draws, an approach to uncertainty estimation distinct from a model’s own stated confidence (Malinin and Gales 2021) – reproduces the null, with answer-saturation itself corroborating decoupling in raw answer-space. Full numbers and caveats: Appendix E.

| config         | before | after | closed | ablation gap [95% CI]       |
|----------------|--------|-------|--------|-----------------------------|
| haiku:std      | 0.357  | 0.127 | 100%   | +0.0016 [+0.0002, +0.0030]* |
| haiku:think    | 0.352  | 0.126 | 100%   | +0.0007 [−0.0008, +0.0022]  |
| sonnet:std     | 0.193  | 0.125 | 102%   | +0.0017 [+0.0002, +0.0031]* |
| sonnet:think   | 0.182  | 0.127 | 100%   | +0.0004 [−0.0007, +0.0017]  |
| opus:std       | 0.186  | 0.126 | 101%   | +0.0003 [−0.0009, +0.0016]  |
| opus:think     | 0.185  | 0.126 | 100%   | +0.0002 [−0.0011, +0.0015]  |
| gemma:std      | 0.178  | 0.127 | 100%   | −0.0005 [−0.0019, +0.0010]  |
| gemma:think    | 0.209  | 0.127 | 100%   | +0.0005 [−0.0010, +0.0019]  |
| deepseek:std   | 0.198  | 0.126 | 101%   | +0.0000 [−0.0009, +0.0010]  |
| deepseek:think | 0.179  | 0.126 | 101%   | +0.0006 [−0.0004, +0.0015]  |

Table 4: Recalibrator PAD before/after and % of the oracle gap (0.127) closed, with the paired ablation gap = PAD(features only) − PAD(features+confidence). \*: CI excludes zero.

| model  | design | before | after | vs. constant   |
|--------|--------|--------|-------|----------------|
| haiku  | A      | 0.356  | 0.168 | worse*         |
| haiku  | B      | 0.356  | 0.147 | tied           |
| haiku  | C      | 0.355  | 0.141 | tied           |
| sonnet | A      | 0.188  | 0.186 | worse*         |
| sonnet | B      | 0.188  | 0.158 | worse*         |
| sonnet | C      | 0.188  | 0.189 | worse*         |
| opus   | A      | 0.184  | 0.189 | worse*         |
| opus   | B      | 0.184  | 0.158 | worse*         |
| opus   | C      | 0.184  | 0.204 | worse* (hurts) |

Table 5: Prompting-intervention table,  $n=300$  held-out, constant PAD= 0.143. \*: gap CI excludes zero; “tied” brackets zero.

| config              | raw   | recalibrated |
|---------------------|-------|--------------|
| haiku:std           | 0.044 | 0.070        |
| haiku:think         | 0.039 | 0.070        |
| sonnet:std          | 0.039 | 0.069        |
| sonnet:think        | 0.052 | 0.070        |
| opus:std            | 0.045 | 0.070        |
| opus:think          | 0.060 | 0.069        |
| gemma:std           | 0.087 | 0.070        |
| gemma:think         | 0.044 | 0.070        |
| deepseek:std        | 0.025 | 0.070        |
| deepseek:think      | 0.050 | 0.070        |
| pooled              | 0.048 | 0.070        |
| ideal ( $c = \pi$ ) |       | 0.028        |

Table 6: Simulated over-reliance, raw vs. recalibrated. Helps only 1/10 configurations.

## 5.12 Over-Reliance: A Second-Order Metric-Design Point

Following Lee and See (2004), a clinician relies on the model’s answer with probability  $\sigma(\beta(c - \theta))$ ; warranted reliance is  $\pi$ ; over-reliance is  $\max(0, P(\text{rely}) - \pi)$ , averaged over  $\theta \in \{.5, .6, .7, .8\}$ ,  $\beta \in \{5, 10, 20\}$ , pooled across all ten configurations. Decoupled confidence gives over-reliance 0.048; ideal ( $c=\pi$ ) gives 0.028 (41% lower). The recalibrator, despite closing the PAD gap for every configuration, lands at **0.070** – worse than raw pooled (45% higher) – because it regresses every configuration toward the oracle’s residual mean, helping only the worst raw configuration (gemma:standard,  $0.087 \rightarrow 0.070$ ) and hurting the other nine, most dramatically the cautious deepseek:standard ( $0.025 \rightarrow 0.070$ ). This ranking holds at every one of the twelve  $(\theta, \beta)$  cells individually. **We read this as a genuine second-order contribution:** PAD-minimization is not over-reliance-minimization – optimizing an average calibration metric can move a safety-relevant quantity the wrong way for models that were already appropriately cautious, a distinction invisible to any purely average-based metric alone.

## 6 Discussion

The picture is confidence tracking something other than clinical difficulty: models respond to features with real power ( $R^2$  up to 0.76) but in contradictory, model-specific ways that align poorly with physician disagreement, suggesting an ar-

chitectural or training-level fix, not an elicitation-level one – neither reasoning nor prompting closes the gap, while a trivial recalibrator does, cheaply, by ignoring confidence almost entirely: useful, but an unflattering diagnosis. Section 5.12 complicates “ship the recalibrator”: a metric-optimal fix is not automatically harm-optimal. The capability trend (Section 5.8) raises, without confirming, a further concern: if more-capable models are more confidently wrong on the hardest cases, scaling capability without fixing the decoupling could worsen over-reliance.

## 7 Limitations and Broader Impact

**Construct validity.** Tiers follow directly from LIDC-IDRI ratings without independent clinical review against radiologists’ own sense of difficulty – the largest open risk here, stated as unresolved. **Single domain.** LIDC-IDRI is our only source; we checked five BI-RADS-style candidates (VinDr-Mammo, CMMD, a Zenodo contrast-enhanced set, an inaccessible 101-radiologist study, CBIS-DDSM), rejecting all as consensus-only or inaccessible per-reader. **Text, not images; discrete  $\pi$ ; DeepSeek manual-only.** No DICOM was available, so models reason over extracted features, not raw CT;  $\pi$ ’s discreteness is mitigated by the tier redefinition and collision-free-subset check, though a continuous source would sharpen every result; DeepSeek is flagged throughout

as manual-collection only. **Theorem 1 is unverified** (Section 4.1). **The pooled correlation is threshold-sensitive by a confirmed mechanism** (Appendix C): concentrated in the  $\approx 61\%$  tier-flipping subset, with a genuine positive signal on the stable  $\approx 39\%$  averaged away in the pooled statistic – a refinement, not an unresolved risk. **Vignette richness.** Only subtlety/spiculation/margin are used; unused LIDC-IDRI fields (calcification, sphericity, lobulation, texture) could shrink the collision rate and raise the oracle target, but we did not re-run the full panel against them. **Availability.** Pipeline code and the derived corpus will be released publicly upon publication under a research-use license.

## A Data Appendix: Example Vignettes and Corpus Statistics

$n=1,413$  nodules, 709 scans (907 four-reader, 506 three-reader). Tiers: high 546, contested 708, ambiguous 159. Feature ranges (1–5 LIDC scale; extent in px): subtlety [1, 5] (mean 3.92), spiculation [1, 5] (mean 1.61), margin [1, 5] (mean 3.99), extent [8.0, 88.6] (mean 21.9). Only 690/1,413 vignettes are unique (Section 3.2). One representative case, exact prompt format: **High** ( $\pi=1.0$ , unanimous, 4 readers): “subtlety 4.50 | spiculation 2.50 | margin 4.00.” A genuine collision from the corpus: cases 581:4 and 581:5 share the identical presented vignette “subtlety 2.25 | spiculation 1.00 | margin 2.75” (both 4 readers), yet 581:4 is ambiguous ( $\pi=0.5$ ) and 581:5 is contested ( $\pi=0.75$ ) – the model sees one prompt for two different agreement targets, illustrating Section 3.2’s caveat with a real corpus pair.

## B Corpus and Matcher Robustness

Matcher stability across tight (20px/2mm), base (30px/3mm, frozen default), and loose (40px/4mm) thresholds: total size swings  $\pm 3\%$  (1,375–1,418); the thinnest tier (ambiguous) moves 152–161 cases ( $\pm 5\%$ ) while remaining essentially the same nodule set (Jaccard  $\geq 0.94$ ). Tier assignment is not a hidden researcher degree of freedom. An exclusion audit shows dropped annotations are  $> 99.9\%$  LIDC’s inherent sub-3mm/uncharacterized marks, not a pipeline artifact.

## C Malignancy-Threshold Sensitivity

$\pi$  depends on a threshold ( $\geq 3$  throughout, the LIDC field convention). Recomputing at the stricter  $\geq 4$  threshold (same 1,413 cases, pure relabeling):  $\pi$  changes for 61.5% of cases, the high tier grows 546 $\rightarrow$ 728, and  $c^*$  moves to 1.0 (PAD 0.162). The baseline result is robust (10/10 still worse than the  $\geq 4$  constant); the pooled correlation is not (e.g., sonnet:standard  $-0.078 \rightarrow +0.271$ ). Three cheap diagnostics, no re-query, resolve why: **(1)** confidence/answer columns are identical across both analyses, so any shift is  $\pi$ -side; **(2)** splitting cases into threshold-stable ( $n \approx 544$ ,  $\pi$  literally identical under both thresholds) vs. threshold-sensitive ( $n \approx 869$ ) shows the shift is entirely in the sensitive subset, where correlation sign flips with the threshold on identical confidences (Table 7), while the stable subset shows a real, invariant positive correlation ( $+0.20$ – $0.25$  for four of five configurations) the pooled statistic averages away; **(3)** MI corroborates this – opus/gemini/deepseek:standard keep significant MI at

| config       | $r$ stable | $r_{\geq 3}$ sens. | $r_{\geq 4}$ sens. |
|--------------|------------|--------------------|--------------------|
| sonnet:std   | +0.247     | −0.294             | +0.286             |
| opus:std     | +0.237     | −0.292             | +0.294             |
| gemini:std   | +0.201     | −0.291             | +0.201             |
| haiku:std    | +0.082     | +0.008             | −0.060             |
| deepseek:std | +0.238     | +0.214             | −0.240             |

Table 7: Correlation decomposition: threshold-stable (single column, identical under both thresholds) vs. threshold-sensitive under each threshold.

both thresholds, while sonnet:standard shows none at either despite its large Pearson swing, confirming that swing is a linear-correlation artifact. **Resolution:** a confirmed, characterized composition effect, not an open mystery; we report the  $\geq 3$  field-standard statistic as primary and treat this as a genuine qualification of the decoupling claim.

## D Capability Trend: Full Results

Standard slope  $+0.067$  (95% CI  $[+0.044, +0.091]$ ); thinking  $+0.095$  ( $[+0.072, +0.119]$ );  $n=159$  shared ambiguous cases, cluster-bootstrapped. Three rungs only, non-monotonic (Sonnet below Haiku on some measures), Opus-driven; strengthening this would need a multi-vendor capability ladder at paid-API scale we did not have – an acknowledged scope limit, not an open action item.

## E Non-Verbalized Confidence: Full Results

Gemini, 10 samples at temperature 0.8 on 73 cases, confidence = fraction agreeing with the modal answer; reproduces the null ( $r = -0.111$ , 95% CI  $[-0.305, +0.145]$ ,  $p=0.35$ ), saturated 0.947–1.00 across tiers – the model gives the same answer nearly every time even on a 50/50 case, corroborating decoupling in raw answer-space. A gpt-oss spot-check (e5-embedded rationale cosine similarity) gives a similarly flat, non-significant result. Caveats: small, single-model-per-variant samples, partial completion ( $n=56$ – $85$ ), free-tier constraints; directionally corroborating, not paper-grade precision.

## Ethical Statement

ContestBench is a measurement instrument, not a diagnostic tool; no evaluated model is validated for clinical use. Our purpose is to surface a failure mode – confidence decoupled from clinical difficulty – that could encourage inappropriate trust where judgment matters most. Two-sided risk: this could be misread as license to distrust LLM confidence uniformly (it is decoupled from this target, not uniformly wrong), or a recalibrator like ours could reduce miscalibration while increasing over-reliance for already-cautious models (Section 5.12). All data is de-identified and public (LIDC-IDRI); no patient data or clinical decisions were involved.

## References

Armato, S. G.; McLennan, G.; Bidaut, L.; McNitt-Gray, M. F.; Meyer, C. R.; Reeves, A. P.; Zhao, B.; Aberle, D. R.;

- Henschke, C. I.; Hoffman, E. A.; et al. 2011. The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A Completed Reference Database of Lung Nodules on CT Scans. *Medical Physics*, 38(2): 915–931.
- Bansal, G.; Wu, T.; Zhou, J.; Fok, R.; Nushi, B.; Kamar, E.; Ribeiro, M. T.; and Weld, D. 2021. Does the Whole Exceed Its Parts? The Effect of AI Explanations on Complementary Team Performance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- Brier, G. W. 1950. Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1): 1–3.
- Buçinca, Z.; Malaya, M. B.; and Gajos, K. Z. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1): 1–21.
- Byun, J.; et al. 2026. Overconfidence and Calibration in Medical Visual Question Answering. arXiv:2604.02543.
- Davani, A. M.; Díaz, M.; and Prabhakaran, V. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics*, 10: 92–110.
- DeGroot, M. H.; and Fienberg, S. E. 1983. The Comparison and Evaluation of Forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1-2): 12–22.
- Desai, S.; and Durrett, G. 2020. Calibration of Pre-Trained Transformers. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Gneiting, T.; and Raftery, A. E. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477): 359–378.
- Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the International Conference on Machine Learning (ICML)*.
- Jiang, Z.; Araki, J.; Ding, H.; and Neubig, G. 2021. How Can We Know When Language Models Know? On the Calibration of Language Models for Question Answering. *Transactions of the Association for Computational Linguistics*, 9: 962–977.
- Jin, D.; Pan, E.; Oufattole, N.; Weng, W.-H.; Fang, H.; and Szolovits, P. 2021. What Disease Does This Patient Have? A Large-Scale Open Domain Question Answering Dataset from Medical Exams. *Applied Sciences*, 11(14): 6421.
- Kadavath, S.; Conerly, T.; Askell, A.; Henighan, T.; Drain, D.; Perez, E.; Schiefer, N.; Dodds, Z. H.; DasSarma, N.; Tran-Johnson, E.; et al. 2022. Language Models (Mostly) Know What They Know. arXiv:2207.05221.
- Kompa, B.; Snoek, J.; and Beam, A. L. 2021. Second Opinion Needed: Communicating Uncertainty in Medical Machine Learning. *npj Digital Medicine*, 4(1): 4.
- Lee, J. D.; and See, K. A. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human Factors*, 46(1): 50–80.
- Li, S. S.; Balachandran, V.; Feng, S.; Ilgen, J.; Pierson, E.; Koh, P. W.; and Tsvetkov, Y. 2024. MEDIQ: Question-Asking LLMs for Adaptive and Reliable Clinical Reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Lin, S.; Hilton, J.; and Evans, O. 2022. Teaching Models to Express Their Uncertainty in Words. *Transactions on Machine Learning Research*.
- Malinin, A.; and Gales, M. 2021. Uncertainty Estimation in Autoregressive Structured Prediction. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Manakul, P.; Liusie, A.; and Gales, M. J. F. 2023. SelfCheck-GPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- MedQADE Authors. 2026. MedQADE: Difficulty-Aware Evaluation of LLM Judges on Open-Response Clinical Question Answering. arXiv:2607.01103.
- MedR-Bench Contributors. 2025. MedR-Bench: A Benchmark for Medical Reasoning. Preprint.
- MedThink-Bench Contributors. 2025. MedThink-Bench: Evaluating Clinical Reasoning in Language Models. Preprint.
- Mielke, S. J.; Szlam, A.; Dinan, E.; and Boureau, Y.-L. 2022. Reducing Conversational Agents’ Overconfidence Through Linguistic Calibration. *Transactions of the Association for Computational Linguistics*, 10: 857–872.
- Murphy, A. H. 1973. A New Vector Partition of the Probability Score. *Journal of Applied Meteorology*, 12(4): 595–600.
- Naeini, M. P.; Cooper, G. F.; and Hauskrecht, M. 2015. Obtaining Well Calibrated Probabilities Using Bayesian Binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Ovadia, Y.; Fertig, E.; Ren, J.; Nado, Z.; Sculley, D.; Nowozin, S.; Dillon, J. V.; Lakshminarayanan, B.; and Snoek, J. 2019. Can You Trust Your Model’s Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Pal, A.; Umapathi, L. K.; and Sankarasubbu, M. 2022. MedMCQA: A Large-Scale Multi-Subject Multi-Choice Dataset for Medical Domain Question Answering. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*.
- Parasuraman, R.; and Riley, V. 1997. Humans and Automation: Use, Misuse, Disuse, Abuse. *Human Factors*, 39(2): 230–253.
- Passi, S.; and Vorvoreanu, M. 2022. Overreliance on AI: Literature Review. Microsoft Research Report.
- Platt, J. C. 1999. Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. *Advances in Large Margin Classifiers*, 10(3): 61–74.
- Savage, T.; et al. 2025. Uncertainty Quantification via Sample Consistency for Large Language Models in Clinical Decision Support. *Journal of the American Medical Informatics Association*.

Singhal, K.; Azizi, S.; Tu, T.; Mahdavi, S. S.; Wei, J.; Chung, H. W.; Scales, N.; Tanwani, A.; Cole-Lewis, H.; Pfohl, S.; et al. 2023. Large Language Models Encode Clinical Knowledge. *Nature*, 620: 172–180.

Steyvers, M.; Tejeda, H.; Kumar, A.; Belem, C.; Karny, S.; Hu, X.; Mayer, L. W.; and Smyth, P. 2025. What Large Language Models Know and What People Think They Know. *Nature Machine Intelligence*, 7: 221–231.

Testoni, A.; and Calixto, I. 2026. Mind the Gap: Specialty-Aware Uncertainty in Clinical Question Answering. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.

Tian, K.; Mitchell, E.; Zhou, A.; Sharma, A.; Rafailov, R.; Yao, H.; Finn, C.; and Manning, C. D. 2023. Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Uma, A. N.; Fornaciari, T.; Hovy, D.; Paun, S.; Plank, B.; and Poesio, M. 2021. Learning from Disagreement: A Survey. *Journal of Artificial Intelligence Research*, 72: 1385–1470.

Wang, X.; Wei, J.; Schuurmans, D.; Le, Q.; Chi, E.; Narang, S.; Chowdhery, A.; and Zhou, D. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Xiong, M.; Hu, Z.; Lu, X.; Li, Y.; Fu, J.; He, J.; and Hooi, B. 2024. Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs. In *Proceedings of the International Conference on Learning Representations (ICLR)*.